FollowNet: A Comprehensive Benchmark for Car-Following Behavior Modeling

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***Abstract*—Car-following is a control process in which a following vehicle (FV) adjusts its acceleration to keep a safe distance from the lead vehicle (LV). Recently, there has been a booming of data-driven models that enable more accurate modeling of car-following through real-world driving datasets. Although there are several public datasets available, their formats are not always consistent, making it challenging to determine the state-of-the-art models and how well a new model performs compared to existing ones. In contrast, research fields such as image recognition and object detection have benchmark datasets like ImageNet, Microsoft COCO, and KITTI. To address this gap and promote the development of microscopic traffic flow modeling, we establish a public benchmark dataset for car-following behavior modeling. The benchmark consists of more than 80K car-following events extracted from five public driving datasets using the same criteria. These events cover diverse situations including different road types, various weather conditions, and mixed traffic flows with autonomous vehicles. Moreover, to give an overview of current progress in car-following modeling, we implemented and tested representative baseline models with the benchmark. Results show that the deep deterministic policy gradient (DDPG) based model performs competitively with a lower MSE for spacing compared to traditional intelligent driver model (IDM) and Gazis-Herman-Rothery (GHR) models, and a smaller collision rate compared to fully connected neural network (NN) and long short-term memory (LSTM) models in most datasets. The established benchmark will provide researchers with consistent data formats and metrics for cross-comparing different car-following models, promoting the development of more accurate models. We open-source our dataset and implementation code in https://github.com/HKUST-DRIVE-AI-LAB/FollowNet.**

***Index Terms*—Car-following, Traffic Simulation, Traffic flow, Datasets, Benchmark.**

I. Introduction

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CAR-FOLLOWING is the most fundamental and frequent driving behavior. It involves actions taken by a driver when following another vehicle ahead. Proper car-following behavior can lower crash risks and improve traffic flow stability [1]–[4]. The corresponding car-following model is a mathematical or computational representation of the behavior exhibited by drivers when following other vehicles on the road. It describes and predicts the dynamics of following vehicles (FV) and lead vehicles (LV) movements in traffic flow and is a cornerstone for microscopic traffic simulation [5], [6].

Over the past decade, there has been a boom of data-driven car-following models, primarily due to the availability of real-world driving data and advancements in machine learning. Representative data-driven car-following models include neural network based [7], [8], recurrent neural network (RNN) based [9], [10], and reinforcement learning (RL) based [11]–[14]. However, existing research has the following limitations:

1) **Lack of standardized data formats and evaluation criterion.** Although there are multiple public driving datasets available (e.g., NGSIM, HighD), there are no standardized car-following datasets and evaluation metrics, making it difficult to compare newly proposed car-following models with existing ones in terms of performance.
2) **Inadequate representation of car-following behavior in mixed traffic flows.** We will likely face a transitional phase of mixed traffic where autonomous and human-driven vehicles share the road. However, previous studies have mainly focused on modeling car-following behavior using limited datasets that do not include autonomous vehicles.

To further advance the field of microscopic traffic simulation modeling, it is imperative to establish a public car-following benchmark that can address the aforementioned issues and serve as a standard dataset for research in car-following. Such a benchmark would be beneficial for advancing the field of microscopic traffic flow modeling, similar to how standard datasets such as ImageNet [15], Microsoft COCO [16], and KITTI [17] have contributed to their respective fields. To achieve this, we have created the benchmark by extracting data from five public datasets based on the same criteria. Our benchmark includes five baseline car-following models, consisting of both traditional and data-driven models (Figure. 1). This paper presents a summary of the recent developments in this field, as well as evaluates the performance of mainstream models using our benchmark.

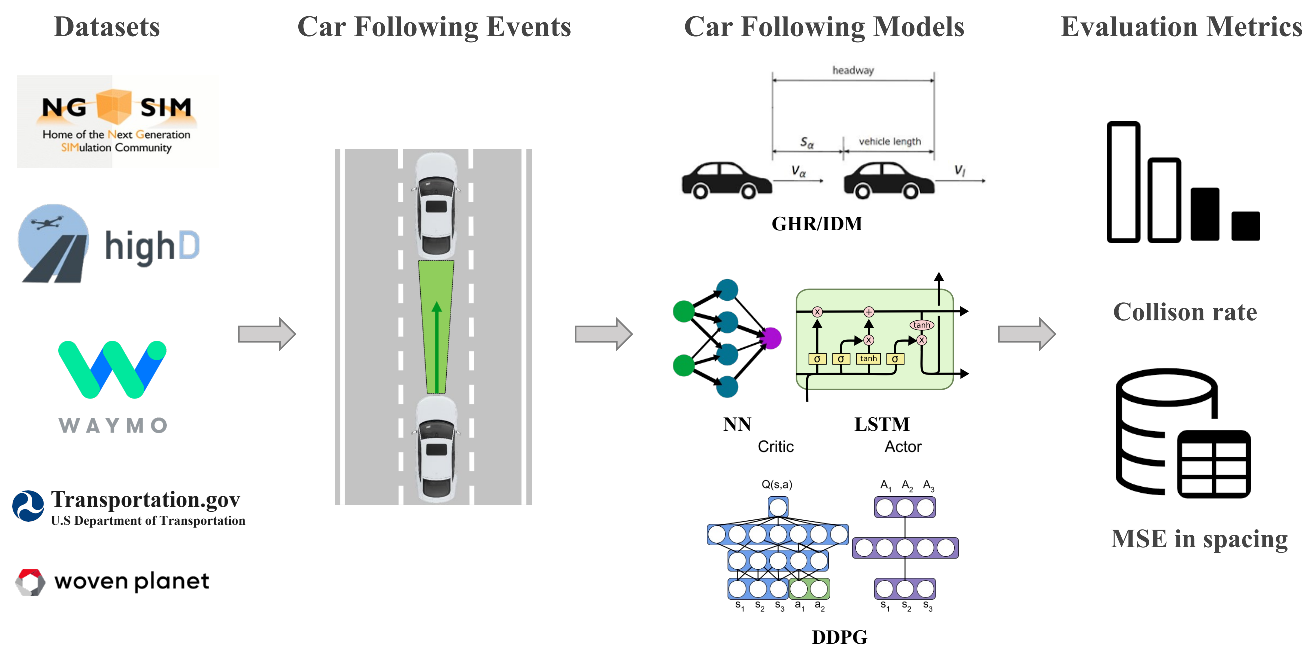

Fig. 1. A Roadmap of Car-following Benchmark: FollowNet

The contributions of this study include:

- Summarized existing car-following models systematically.
- Established the first benchmark of the car-following behavior to save the effort of data extraction and facilitate the development of car-following models.
- Offered diverse scenarios for testing car-following models, including mixed traffic and different road types, while evaluating these models using standardized metrics.
- Provided references for the community on car-following extraction, calibration, and baseline model implementations, through data and code open-sourcing.

The rest of the paper is organized as follows. Section II reviews car-following models. Section III analyzes the characteristics of car-following events in different datasets. Section IV introduces five baseline models. Section V evaluates the model performance with consistent metrics. Section VI discusses future directions of car-following research. Section VII concludes the paper.

II. Related Work

A. Traditional Car-Following Models

Since the Pipes model was proposed in 1953 [18], researchers have studied car-following models for 70 years. In general, the existing traditional car-following models are divided into five main categories: stimulus-response model, safe distance model, psycho-physiological model, optimal velocity model, and desired goal model [19]–[21].

The stimulus-response model uses the relative distance or relative speed to determine the acceleration of the FV. The Gazis-Herman-Rothery (GHR) model [22], the first stimulus-response model, assumes that a vehicle's acceleration positively relates to the speed difference and that the FV's response is expressed regarding its acceleration or deceleration behavior. The basic equation of this model can be expressed as:

$$a_n(t) = c v_n^m(t) \frac{\Delta v(t-T)}{\Delta x^l(t-T)} \tag{1}$$

where the n-th vehicle's acceleration at time t is represented by $a_n(t)$, Δv, and Δx are the speed difference and distance between the $(n-1)$ th vehicle and the n th vehicle, respectively. m, c, and l are constants that need to be calculated, and T is the driver's reaction time. The GM car-following model [23] is another well-known stimulus-response model which used the difference between the LV and FV along with the velocity and space headway to determine the acceleration.

[24] proposed the first safety distance model. The underlying principle of the model differs from the stimulus-response model due to the LV's unpredictable motion. The model is also known as the conflict avoidance model since the FV keeps a minimum safe distance even though the LV implements sudden breaks. Its basic formula is:

$$\Delta x(t-T) = \alpha v_{n-1}^2(t-T) + \beta_l v_n^2(t) + \beta v_n(t) + b_0 \tag{2}$$

where Δx denotes the distance between the $(n-1)$th vehicle and the n-th vehicle, $v_n(t)$ is the speed of the n-th vehicle at time t, T is the driver's reaction time, α, β_l, β, b_0 are parameters to be calibrated. Similarly, Gipps model [5] utilizes the concept of the safe distance to facilitate car-following behavior. Specifically, the driver of the FV selects a certain speed and maintains a corresponding distance from the LV to prevent a collision.

The psycho-physiological model suggests that drivers adopt strategies based on the relative motion between the LV and FV, including changes in speed and distance differences, and only react when the threshold value is exceeded. This model was